

Programs of Studies, School Modality

NAME OF THE SUBJECT: **COMPUTATIONAL TOOLS AND ALGORITHMS**

AREA:

CODE: **COM-16301**

GENERAL OBJECTIVE(S) OF THE SUBJECT:

Students recognizes the importance of the computational tools for professional and academic work, solving problems through a process of calculation, graphic analysis and programming of algorithms.

Students develops ability to solve problems, analyze and present information and model real situations applying reasoning, notions of mathematics and computational tools.

The student acquires the skills to work with at least one scripting language, knowing its advantages and limitations.

TOPICS AND SUBTOPICS:

- I. Introduction.
 - 1.1 Presentation of work and evaluation method.
 - 1.2 Background and vision of information and communications technology, the computing process and programming languages, electronic devices and sensors, communications networks: Internet/ Internet of things.
- II. Computational calculation and programming in the computing environment.
 - 2.1 Introduction to scripting languages. Python, characteristics, generalities and operation. Code in scripts, execution of instructions, types of values, constants and variables. Numerical expressions and operators: arithmetic, relational and logical. Predefines constants, predefined functions of the numerical or mathematical libraries.
 - 2.2 Basic iterable structures: lists and references, sequences of integer values, set and texts. Indexing, modifications or changes. Operations and manipulation with functions and methods. Transformation of lists and sets. Set operations: union, intersection and difference. Textual manipulations and analysis.
 - 2.3 Visualization of data and graphs in the plane, line, dispersion, histograms, bars and pie charts. Additional elements of graphics: labels, types of lines, symbols and colors. Graphic libraries.
 - 2.4 Notion of algorithms. Solutions in pseudocode. Graphic representation of algorithms by means of flowcharts. Data entry and display of results.
 - 2.5 Syntax of programs. Editing, protection, recovery and execution of programs.
 - 2.6 Definition and application of functions.
 - 2.7 Preparation and import of libraries.
 - 2.8 Programming instructions.
 - 2.9 Identification of levels to subordinate instructions.
 - 2.10 Reading text files, reading per element, reading by line. Data separation.
 - 2.11 Use of dictionaries to organize data by keys.
 - 2.12 Use of pandas dataframes to work tables in an integrated manner.
 - 2.13 Reading of data files to dataframe and writing of dataframes to data files.
 - 2.14 Analytical facilities offered by dataframes. Record selection.
 - 2.15 Calculations with values in columns. Creation of new columns.
 - 2.16 Graphical data facilities in data frames.
- III. Analysis of information to solve problems with the electronic sheet (Excel).
 - 3.1 Introduction: menus and tools, dialog Windows, cells, sheets and books. Text and numerical data.
 - 3.2 Formulas and references to the cells. Relative cell references, absolute cell references.

- 3.3 Use of predefined arithmetic, trigonometric and data/time functions. Creation and edition of several graphs from data tables.
- 3.4 Definition of tables. Searches and extraction of table record values.
- 3.5 Use of dynamic and pivot tables to summarize data.

LEARNING ACTIVITIES:

Computational Tools and Algorithms is a course that provide the student with an introduction to the use of the computer as a tool. The learning is conducted by the teacher with computational resources and practical elements of three types: examples, exercises and problems that are presented to students as work in and out of class with tasks and projects.

The work environment allows the student to solve problems through numerical calculations, graphs and algorithms programming (Python language and Excel or Open office electronic spreadsheet). The student uses the computer system, which includes hardware and software, as a tool for analysis and problem solving that not only automates the mechanical aspect of solving a problem. Not only facilitating the execution of mathematical operations and data analysis and helping an adequate presentation of the results, but also allowing a process of systematic work of analysis and synthesis interactive and progressive.

The course material is displayed in digital or electronic files prepared for each of the computer work environments. The problems exposed and the level of complexity involved start from basic concepts that are applied and validated by the practical elements mentioned (examples, exercises and problems).

The working method is based on proposing to the student interesting practical problems, which are treated on a scheme of four essential stages: 1) understand the problem, 2) design a solution to the problem, 3) implement the solution with the help of the computer and 4) validate the solution and the results. The teacher serves as a guide, whose role should be focused on motivating the analysis of the problems and guiding the solutions by asking questions that are not necessarily specific to the problem. The teacher must conduct a process of questioning, assimilation and understanding of the problem, as well as provoke the discovery, construction and validation of the solution and the analysis of the results.

In this course the student stimulates and develops his abilities of analysis of problems and synthesis of his solutions, as much with individual work as team work, central aptitudes of students and professionals of diverse disciplines like applied mathematics, economy, actuary, engineering, administration and accounting, among others.

EVALUATION OF THE COURSE:

Final Grade[†]= 30% midterm exam No. 1 + 30% midterm exam No. 2 + 30% Final exam + 10% homework.

Note: is required to pass the final exam to pass the course.

BIBLIOGRAPHY:

The course does not require a textbook because the material is presented in digital or electronic files prepared for each of the computer work environments, but the following texts can support the learning process.

- Downey, A.B., *Think Python: how to think like a computer scientist*, O'Reilly Media; version 2.0.17, January 2016. (3/8/2017 <http://greenteapress.com/thinkpython/thinkpython.pdf>)
- Walkenbach, J., *Microsoft® Excel® 2016 bible*, Indianapolis, Ind.: John Wiley & Sons, c2016.
- Larsen, R.W., *Engineering with Excel* (5th Edition)/ Ronald W. Larsen.--5th ed.-- Upper Saddle River, N. J.: Pearson, 2017, ISBN-13: 9780134589664, ISBN-10: 0134589661
<https://www.pearson.com/us/higher-education/program/Larsen-Engineering-with-Excel-5th-Edition/PGM1101118.html?tab=order>.